



Shan-Cheng Chang 張善程

AI Application Engineer · Document AI / LLM Agents / Knowledge Graph

AI application engineer with 2 years of shipping production LLM systems in manufacturing. At BenQ Materials I built the company's document-AI stack from zero — six OCR + LLM systems now live across finance, procurement, HR and customs — and then moved up the stack to an enterprise knowledge graph, a Claude Code-style agent harness, and an AI data-analysis platform for a polarizer production line. I own the full loop from user interviews and architecture to deployment and adoption. M.S. research in AIoT and computer vision (journal paper, conference paper, patent).

6

AI systems in production

97%+

avg. OCR field accuracy

1,500+ h

saved per year

~1M NTD

annual cost benefit

EXPERIENCE

AI Application Engineer

Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, AI Project Office (DCA1) · Taoyuan, Taiwan

- **Built the company's document-AI platform from scratch** (Azure Document Intelligence + Azure OpenAI GPT-4o / o3 / o4-mini, Python Flask & FastAPI, React, SQL Server, IIS). Defined the reusable OCR → LLM extraction → validation → feedback pipeline that every later project was cloned from.
- **Quotation OCR** (live May 2025): raised field accuracy from ~51% to **100%** and cut processing time 3x; 758+ quotations processed; automated procurement audit pre-check; multi-format (PDF/image/Office) and multi-language (ZH/EN/JA) inputs; explainable results via highlighted, zoomable bounding boxes.
- **Taiwan Invoice OCR** (live Dec 2025): 6 invoice formats incl. handwritten and thermal-paper; **97%** accuracy (+19 pp); saves **850 h/yr (595K NTD)** and avoids 480–760K NTD/yr of outsourcing.
- **BMS (China) Invoice OCR**: 11 simplified-Chinese fields at **100%** accuracy; 597 h/yr (305K NTD); rolled out to BMS, DTB and BMM sites for vendor, procurement and finance teams.
- **New-hire Health-Check OCR Platform** (live Sep 2025): 48 medical fields at **97.4%**; FastAPI + React system with AES-256-GCM field-level encryption (ISO 27001), AD/LDAP + JWT auth, 3-tier RBAC, SSE live progress, audit logging and quarterly data purge with deletion evidence; 216+ reports across BMC/BMY/BML sites.
- **Customs OCR**: unified 6 vendor formats (PDF + Excel) with container-number mapping logic, 85% accuracy, reducing re-keying for customs declarations.
- **Supplier ESG Questionnaire System** (Oct 2025 – Mar 2026, first self-built GCP service): React 19 + Express 5 on Cloud Functions + Firestore + Firebase Hosting; Google Forms/Sheets two-way sync, automatic reminders, RBAC; Microsoft Entra ID login saved ~10K NTD per user (~200K NTD). 33K+ LOC, 52 endpoints, delivered solo in 4.5 months using AI-assisted development.

CONTACT

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CORE SKILLS

LLM & Agents

Prompt Engineering RAG
Knowledge Graph LLM Agents
Harness Engineering MCP
Azure OpenAI Gemini API

Document AI / CV

Azure Document Intelligence
OCR + LLM Extraction OpenCV
YOLOv8 Mask R-CNN PyTorch

Data / ML

XGBoost SHAP Neo4j Qdrant

Backend & Web

Python FastAPI Node.js / Express
React TypeScript SQL Server
PostgreSQL Firestore
REST / GraphQL / SSE

Cloud & DevOps

Docker
GCP (Firebase, Cloud Functions, Cloud Run)
Azure AI Git

Security

AES-256-GCM AD / LDAP
Entra ID OAuth ISO 27001 audit logs

EDUCATION

National Taiwan University

M.S. Biomechanics Eng. (AIoT / CV) · 2021 – 2023

National Taipei University of Technology

B.S. Mechanical Engineering · 2017 – 2021

LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2
Military service completed

- **Enterprise Brain → AI Data Analysis Platform** (Dec 2025 – present, lead developer): started as two PoCs — a knowledge graph (Neo4j + PostgreSQL + vector search, validated on cosmetics-regulation compliance, with review gate and versioning) and a Claude Code-style agent harness (Gemini function calling, Memory / Tool / Sandbox / Cron, prompt version control) — then productized as the on-prem AI platform for the M4 polarizer line: Skill/Bash/Read tools, MCP, sandboxed execution, append-only audit log; 6 n8n workflows migrated into 1 skill (7 analysis methods) + 7 flows; XGBoost yield model (50 sensors, F1 0.74) with SHAP recipe cards; sensor data never leaves the intranet.
- **Prompt governance**: designed the APO-OCR architecture (golden set + pure-code validator + Coach LLM + budget/loop circuit breakers) and mentored an intern to ship it — a 2-iteration, \$1.44 run produced a prompt that scored 8/8 on a held-out set and was 35% shorter than the hand-tuned version.
- Maintained the ESG carbon & energy platform (zero-downtime server migration); TA for two ChatGPT Team courses (67 trainees).

Shift Supervisor (值班股長)

Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- Ran shift operations on the cold-rolled coil line and resolved production anomalies, sustaining continuous production at 95% yield.
- Used process parameters and quality data for root-cause analysis of coil defects and proposed process adjustments to cut rework and downtime; coordinated upstream/downstream lines and QA to meet customer orders.
- Manufacturing and process-data experience that later grounded the production-line AI analytics platform (sensor analysis, SPC, yield modelling) at BenQ Materials.

RESEARCH & SELECTED PROJECTS

AIoT Pest Monitoring & Bee-Pollination Optimization (M.S. thesis, NTU)

2021 – 2023

Lead developer of greenhouse AIoT systems: YOLOv8 + StrongSORT bee tracking (99.5% mAP, 97.7% tracking accuracy), Mask R-CNN (Swin-T) pollen segmentation, deep-learning pest counting (95.3% mAP, 95.8% count accuracy) and kriging-based climate control.

PyTorch · YOLOv8 · StrongSORT · Mask R-CNN · IoT sensors · Kriging

Universal OCR Platform — no-code OCR workflow builder (side project)

2025 – 2026

Self-serve platform that generalizes the company OCR know-how: upload documents, define fields and prompts, run a 7-step pipeline (preprocess → Document Intelligence → multi-attempt LLM extraction → selection → positioning → annotation → validation) and deploy it as a REST API. Multi-file joint extraction with source attribution; three deployment variants (PostgreSQL + SSE, Firestore + Firebase, Supabase Realtime + Cloud Run).

FastAPI · React 19 · PostgreSQL / Supabase · Cloud Run · Firebase Hosting · Azure OpenAI / Gemini · Claude MCP · Docker

Autonomous Navigation with Autoware / ROS (B.S. capstone, NTUT)

2020 – 2021

Indoor/outdoor self-driving platform on Ubuntu + ROS + Autoware (mapping, localization, path planning) with an ML mask-detection module.

PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665.

Chang, S.-C., et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB*.

